

Ahmed Abdulla

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Education

Cornell University, Bachelor of Science | **GPA: 3.7/4.0**

Expected May 2027

- **Major:** Information Science **Minor:** Computer Science
- **Coursework:** Operating Systems, Systems Architecture, Databases, Data Structures & Algorithms, Machine Learning, Functional Programming, Object-Oriented Programming, Probability & Statistics, Linear Algebra, Discrete Math
- **Honors:** Dean's List, 1st Place – Cornell Hackathon

Work Experience

Research Software Engineer | **Cornell Lab of Ornithology** | **Ithaca, NY**

Sep 2025 – Present

- Engineered distributed data pipelines merging drone orthomosaics and acoustic recordings using Python, FastAPI, and PostgreSQL, boosting data throughput by 45%.
- Optimized CNN and Transformer-based species detection models with GPU-accelerated PyTorch training for real-time inference.
- Deployed containerized retraining workflows on Google Cloud with Docker, Prefect, and CI/CD pipelines for automated monitoring and updates.

Software Engineering Intern | **Simulacrum Inc.** | **New York, NY**

May 2025 – Aug 2025

- Built and deployed FastAPI microservices for time-series forecasting, handling **5M+ requests/month** with sub-150ms latency.
- Optimized throughput using async I/O, Redis caching, and containerized deployments via Docker & Nginx on Vertex AI.
- Developed TypeScript SDK and CI/CD pipelines with GitHub Actions & Prometheus, reducing integration time by **35%**.

Software Engineering Intern | **Iconectiv LLC** | **Bridgewater, NJ**

May 2024 – Aug 2024

- Developed Python ETL pipelines to clean and load 14M+ GNS records into Oracle SQL, reducing runtime by 35%.
- Built an Angular/PrimeNG dashboard enabling cross-departmental access to processed data in real time.
- Automated quarterly refresh and validation tasks using Bash, improving consistency and auditability by 40%.

Leadership & Involvements

Cornell Data Strategy | **Vice President & Director of Technology**

Aug 2024 – Present

- Led 25+ developers across 3 pods delivering 4+ production-grade data platforms for enterprise clients (600M+ scale).
- Standardized CI/CD pipelines and on-call rotations, cutting incident resolution time by 40%.
- Architected backend infrastructure and guided developers through code reviews, version control, and deployment best practices.

Publications

Goktas, D., Riaño-Briceño, G., Abdulla, A., et al. (2025). *TempusBench: An Evaluation Framework for Time-Series Forecasting*. **NeurIPS 2025 Workshop on Recent Advances in Time Series Foundation Models (BERT2S). [OpenReview](#)**

Technical Projects

JobLink – AI Job Application Tracker | **Flask, PostgreSQL, scikit-learn, spaCy, OpenAI LLM**

Apr 2025

- Developed secure full-stack platform to centralize job application data with REST APIs and a responsive frontend dashboard.
- Implemented NLP classification pipeline using TF-IDF, RandomForest, and OpenAI LLM verification — achieving 97.8% precision in email entity extraction.
- Deployed on AWS Lambda with CI/CD workflows, automated retraining, and secure PostgreSQL backend.

Poultry Health Monitoring Platform – 1st Place, Cornell Hackathon | **PyTorch, TensorFlow, CUDA**

Feb 2025

- Engineered multimodal AI pipeline integrating image & acoustic data for real-time poultry disease detection with 92% precision.
- Built GPU-optimized CNN models and audio classifiers with PyTorch & TensorFlow, deployed with FastAPI.
- Implemented real-time monitoring and alert dashboard improving farm decision response time by 30%.

Technical Skills

- **Languages:** Python, Java, C/C++, JavaScript, TypeScript, SQL, OCaml, Assembly, HTML/CSS
- **Frameworks/Tools:** React, Spring Boot, Flask, FastAPI, Docker, Kubernetes, Nginx, Jenkins, Jira, Redis
- **Databases/Cloud:** PostgreSQL, MySQL, DynamoDB, Firebase, AWS (S3, EC2, Lambda, CloudWatch), Vertex AI
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Transformers, XGBoost, LSTM, Pandas, NumPy